Exam Overview

The AP Chemistry Exam assesses student application of the science practices and understanding of the learning objectives outlined in the course framework. The exam is 3 hours and 15 minutes long and includes 60 multiple-choice questions and 7 free-response questions. A scientific or graphing calculator is recommended for use on ***both sections*** *of the exam. Students are provided with the periodic table and a formula sheet that lists specific and relevant formulas for use on the exam (see Appendixes). The details of the exam, including exam weighting and timing, can be found below:*

Section	Question Type	Number of Questions	Exam Weighting	Timing
I	**Multiple-choice questions**	**60**	**50%**	**90 minutes**
II	**Free-response questions**		**50%**	**105 minutes**
	Long questions (10 points each)	3		
	Short questions (4 points each)	4		

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The AP Exam assesses the nine units of the course with the following exam weighting on the multiple-choice section:

Units of Instruction	Exam Weighting
Unit 1: Atomic Structure and Properties	**7–9%**
Unit 2: Compound Structure and Properties	**7–9%**
Unit 3: Properties of Substances and Mixtures	**18–22%**
Unit 4: Chemical Reactions	**7–9%**
Unit 5: Kinetics	**7–9%**
Unit 6: Thermochemistry	**7–9%**
Unit 7: Principles of Equilibrium	**7–9%**
Unit 8: Acids and Bases	**11–15%**
Unit 9: Thermodynamics and Electrochemistry	**7–9%**

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How Student Learning Is Assessed on the AP Exam

Section I: Multiple-Choice

Science Practices 1, 2, 4, 5, and 6 are all assessed in the multiple-choice section with the following exam weighting (Science Practice 3 is not assessed in the multiple-choice section):

Science Practice	Exam Weighting
Practice 1: Models and Representations	**8–12%**
Practice 2: Question and Method	**8–12%**
Practice 4: Model Analysis	**23–30%**
Practice 5: Mathematical Routines	**35–42%**
Practice 6: Argumentation	**8–12%**

Section II: Free-Response

All six science practices are assessed in the free-response section with the following exam weighting:

Science Practice	Exam Weighting
Practice 1: Models and Representations	**2–4%**
Practice 2: Question and Method	**10–16%**
Practice 3: Representing Data and Phenomena	**8–16%**
Practice 4: Model Analysis	**5–9%**
Practice 5: Mathematical Routines	**43–53%**
Practice 6: Argumentation	**15–24%**

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Task Verbs Used in Free-Response Questions

The following task verbs are commonly used in the free-response questions:

Calculate: Perform mathematical steps to arrive at a final answer, including algebraic expressions, properly substituted numbers, and correct labeling of units and significant figures.

Describe: Provide the relevant characteristics of a specified topic.

Determine: Make a decision or arrive at a conclusion after reasoning, observation, or applying mathematical routines (calculations).

Estimate: Roughly calculate numerical quantities, values (greater than, equal to, less than), or signs (negative, positive) of quantities based on experimental evidence or provided data.

Explain: Provide information about how or why a relationship, process, pattern, position, situation, or outcome occurs, using evidence and/or reasoning to support or qualify a claim. Explain "how" typically requires analyzing the relationship, process, pattern, position, situation, or outcome; whereas, explain "why" typically requires analysis of motivations or reasons for the relationship, process, pattern, position, situation, or outcome. Also phrased as "give one reason."

Identify/Indicate/Circle: Indicate or provide information about a specified topic in words or by circling given information. Also phrased as "What is?" or "Which?" or other interrogatory words.

Justify: Provide evidence to support, qualify, or defend a claim and/or provide reasoning to explain how that evidence supports or qualifies the claim.

Make a claim: Make an assertion that is based on evidence or knowledge.

Predict/Make a prediction: Predict the causes or effects of a change in, or disruption to, one or more components in a relationship, pattern, process, or system.

Represent/Draw/Write an Equation/Complete a Diagram: Use appropriate graphs, symbols, words, and/or models to describe phenomena, characteristics, and/or relationships.

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Sample Exam Questions

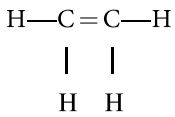
The sample exam questions that follow illustrate the relationship between the course framework and AP Chemistry Exam and serve as examples of the types of questions that appear on the exam. After the sample questions is a table that shows which skill, learning objective(s), and unit each question relates to. The table also provides the answers to the multiple-choice questions.

Section I: Multiple-Choice

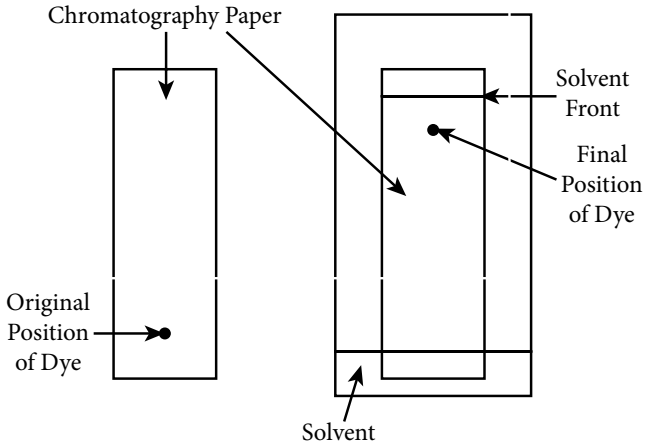
The following are examples of the kinds of multiple-choice questions found on the exam.

$$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$$

1. Which of the following species has the electron configuration shown above?
   (A) O
   (B) Ne
   (C) K^+
   (D) $C1^+$



2. A Lewis diagram for the molecule C_2H_4 is shown above. In the actual C_2H_4 molecule, the H-C-H bond angles are closest to
   (A) 90°
   (B) 109.5°
   (C) 120°
   (D) 180°



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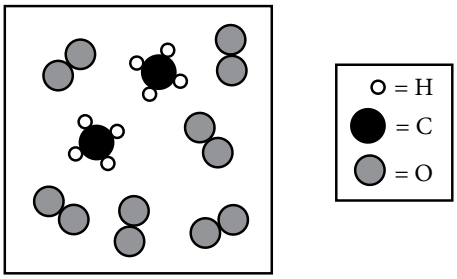
3. Based on the results of the paper chromatography experiment shown above, which of the following can be concluded about the dye?

(A) It has a small molar mass.

(B) It has weak intermolecular forces.

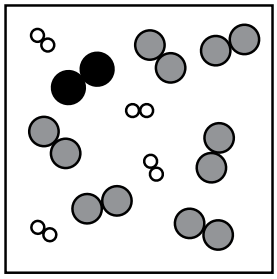
(C) It has a weaker attraction for the stationary phase than it has for the mobile phase.

(D) It has a stronger attraction for the stationary phase than it has for the mobile phase.

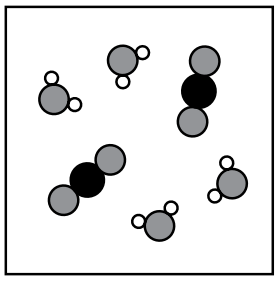


4. The reactants represented above are placed in a vessel and a reaction occurs. Which of the following best represents the contents of the vessel after the reaction has proceeded as completely as possible?

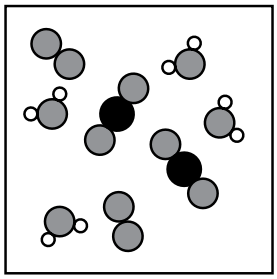
(A)



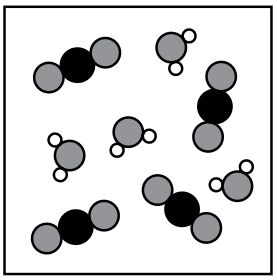
(B)



(C)



(D)



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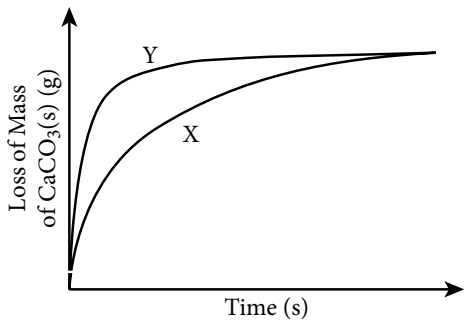
Initial buret reading	2.17 mL
Final buret reading	22.17 mL

5. An antacid tablet containing $Mg(OH)_2(s)$ (molar mass 58.3 g/mol) is titrated with a 0.100 *M* solution of $HNO_3(aq)$. The end point is determined by using an indicator. Based on the data in the table above, what was the mass of the $Mg(OH)_2(s)$ in the antacid tablet?
   (A) 0.0583 g
   (B) 0.583 g
   (C) 5.83 g
   (D) 58.3 g

$$HCl(aq) + CaCO_3(s) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$$

The reaction between $HCl(aq)$ and $CaCO_3(s)$ is represented by the equation above. Two separate trials were carried out using $CaCO_3(s)$ samples of the same mass, but one sample was a single piece of $CaCO_3(s)$, and one sample was composed of small pieces of $CaCO_3(s)$. The loss of mass of $CaCO_3(s)$ as a function of time for both trials is shown in the graph below.



6. Which of the curves, X or Y, represents the reaction with small pieces of $CaCO_3(s)$, and why?
   (A) Curve X, because it shows that the reaction proceeded at a uniform rate.
   (B) Curve X, because it takes a shorter time for the reaction to go to completion due to the larger surface area of $CaCO_3(s)$.
   (C) Curve Y, because it shows that the reaction proceeded at a nonuniform rate.
   (D) Curve Y, because it takes a shorter time for the reaction to go to completion due to the larger surface area of $CaCO_3(s)$.

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Cup	Material of Spoon	Initial Temperature of Spoon (°C)	Mass of Spoon (g)	Specific Heat Capacity (J/(g · °C))
A	Aluminum	20	10.0	0.90
B	Ceramic	20	10.0	0.80
C	Steel	20	20.0	0.45
D	Silver	20	40.0	0.23

7. Four identical 50 mL cups of coffee, originally at 95°C, were stirred with four different spoons, as listed in the table above. In which cup will the temperature of the coffee be highest at thermal equilibrium? (Assume that the heat lost to the surroundings is negligible.)

   (A) Cup A

   (B) Cup B

   (C) Cup C

   (D) Cup D

$$Ag^+(aq) + 2\ NH_3\ (aq) \rightleftarrows [Ag\ (NH_3)_2]^+\ (aq)\ K = 1 \times 10^7$$

8. Equal volumes of 0.1 *M* $AgNO_3(aq)$ and 0.4 *M* $NH_3(aq)$ are mixed and allowed to reach the equilibrium represented above. Which of the following correctly lists the equilibrium concentrations of the $Ag^+(aq)$, $NH_3(aq)$, and $[Ag(NH_3)_2]^+(aq)$ in order from least to greatest?

   (A) $[Ag^+]_{eq} < [NH_3]_{eq} < [[Ag(NH_3)_2]^+]_{eq}$

   (B) $[NH_3]_{eq} < [Ag^+]_{eq} < [[Ag(NH_3)_2]^+]_{eq}$

   (C) $[Ag(NH_3)_2]^+]_{eq} < [Ag^+]_{eq} < [NH_3]_{eq}$

   (D) $[Ag^+]_{eq} < [[Ag(NH_3)_2]^+]_{eq} < [NH_3]_{eq}$

$$N_2(g) + 3\ H_2(g) \rightleftarrows 2\ NH_3(g)$$

A 2.0 L reaction vessel contains an equilibrium mixture of the system represented above. The partial pressures of the components of the mixture at equilibrium are listed in the table below.

P_{N_2}	P_{H_2}	P_{NH_3}
0.30 atm	0.50 atm	0.20 atm

9. Which of the following is the best estimate of the total pressure of the system when the volume of the container is reduced to 1.0 L at a constant temperature?

   (A) $P_{total} = 1.00$ atm

   (B) $1.00\ \text{atm} < P_{total} < 2.00\ \text{atm}$

   (C) $P_{total} = 2.00$ atm

   (D) $P_{total} > 2.00$ atm

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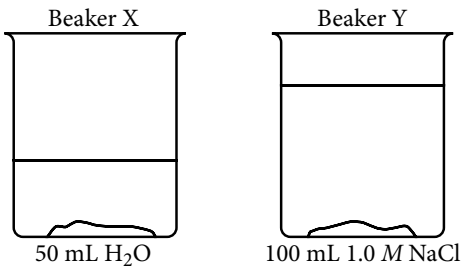

10. Beaker X contains 50 mL of distilled water and beaker Y contains 100 mL of 1.0 *M* NaCl. Solid AgCl is added to each of the beakers. After thoroughly stirring the contents of the beakers, some solid AgCl remains at the bottom of each beaker, as shown above. Which of the following is true?
    (A) $[Ag^+]$ is zero in both beakers.
    (B) $[Ag^+]$ is the same, but not zero, in both beakers.
    (C) $[Ag^+]$ is greater in beaker X.
    (D) $[Ag^+]$ is greater in beaker Y.

11. The value of K_w for water at 0°C is 1×10^{-15}. What is the pOH of water at 0°C?
    (A) 6.5
    (B) 7.0
    (C) 7.5
    (D) 8.0

Half-Reaction	$E°$(V)
$Ag^+(aq) + e^- \rightarrow Ag(s)$	0.80
$Cr^{3+}(aq) + 3\,e^- \rightarrow Cr(s)$	−0.41

12. Based on the standard reduction potentials in the table above, what is the value of $E°$ for a standard galvanic cell made with Ag/Ag^+ and Cr/Cr^{3+} half-cells?
    (A) 0.39 V
    (B) 1.21 V
    (C) 1.99 V
    (D) 2.81 V

Questions 13–15 refer to the information below.

A 0.50 g sample of Mg(*s*) was placed in a solution of HCl(*aq*), where it reacted completely.

13. Which of the following equations best represents the reacting species in the reaction described above?
    (A) $Mg(s) + 2\,H^+(aq) \rightarrow Mg^{2+}(aq) + H_2(g)$
    (B) $Mg(s) + HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$
    (C) $Mg(s) + 2\,HCl(aq) \rightarrow MgCl_2(s) + 2\,H^+(aq)$
    (D) $Mg(s) + 2\,HCl(aq) \rightarrow Mg^{2+}(aq) + Cl_2(g) + H_2(g)$

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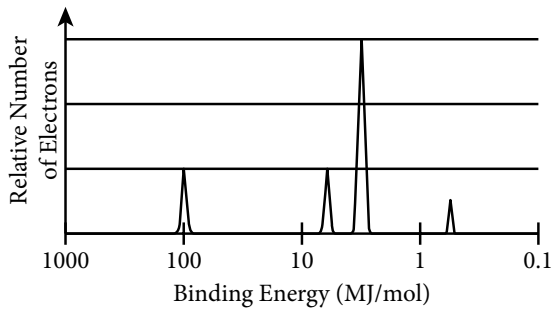
14. In another experiment, a small piece of Mg(*s*) is weighed, then placed in a flask containing excess 1 *M* HCl(*aq*). The student wants to determine the number of moles of gas produced. Which of the following is the best way to conduct the experiment for accurate data collection?
    (A) Conducting the experiment at different temperatures to see which generates the most gas
    (B) Completing the entire reaction in a large Erlenmeyer flask of known volume to measure the volume of the gas collected
    (C) Collecting the gas in a eudiometer tube and measuring the volume of the gas collected
    (D) Conducting the reaction in a graduated cylinder and measuring the volume of the gas collected

15. In a third experiment, 0.10 g samples of Mg(*s*) are placed in excess HCl(*aq*) of various concentrations: 0.050 *M*, 0.10 *M*, 0.25 *M*, and 0.50 *M*. The reactions are run in successive order from 0.050 *M* to 0.50 *M*, and the time required for each reaction to go to completion is recorded. As the concentration of HCl(*aq*) increases from 0.050 *M* to 0.50 *M*, which of the following is the expected result?
    (A) The reaction time increases, and the rate of the reaction decreases.
    (B) The reaction time decreases, and the rate of the reaction increases.
    (C) Both the reaction time and the rate of the reaction increase.
    (D) Both the reaction time and the rate of the reaction decrease.

Section II: Free-Response

The following are examples of the kinds of free-response questions found on the exam. Note that on the actual AP Exam, there will be three long free-response questions and four short free-response questions.



1. The complete photoelectron spectrum of an unknown element is given above.
    (a) Draw an X above the peak that corresponds to the orbital with electrons that are, on average, closest to the nucleus. Justify your answer in terms of Coulomb's law.
    (b) Based on the spectrum, write the complete electron configuration of the element.
    (c) On the graph, draw the peak(s) corresponding to the valence electrons of the element that has one more proton in its nucleus than the unknown element has.

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2. The following questions relate to sulfur and some of its compounds.

(a) Write the balanced equation for the combustion of $S_8(s)$ to form $SO_2(g)$.

(b) Calculate the volume of $O_2(g)$, measured at 1.00 atm and 298 K, that is required to completely combust a 500.0 g sample of pure $S_8(s)$.

(c) A student claims that the combustion of S_8 is an oxidation-reduction reaction. Justify the claim by identifying the oxidation numbers of sulfur and oxygen both before and after the reaction.

(d) In the box below, draw a Lewis electron-dot diagram for one valid resonance structure of SO_2.

(e) Based on the diagram you drew in part (d), what is the approximate oxygen-sulfur-oxygen bond angle in SO_2?

SO_2 can be oxidized to form SO_3 according to the following equation.

$$2\ SO_2(g) + O_2(g) \rightarrow 2\ SO_3(g) \quad \Delta H° = -198\ \text{kJ/mol}_{rxn}$$

(f) Is the value of $\Delta S°$ for the reaction represented above positive or negative? Justify your answer.

$$2\ SO_2(g) + O_2(g) \rightarrow 2\ SO_3(l)$$

(g) Is the magnitude of $\Delta H°$ for the reaction to form $SO_3(l)$, represented above, greater than, less than, or equal to the magnitude of $\Delta H°$ for the reaction to form $SO_3(g)$? Justify your answer.

(h) Based on the information above, how does the thermodynamic favorability of the reaction change as the temperature of the reaction system is decreased? Justify your answer.

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Answer Key and Question Alignment to Course Framework

Multiple-Choice Question	Answer	Skill	Learning Objective	Essential Knowledge
1	C	1.A	1.5.A	1.5.A.3
2	C	6.C	2.7.A	2.7.A.2
3	C	2.D	3.9.A	3.9.A.1
4	C	5.C	4.5.A	4.5.A.1
5	A	5.F	4.6.A	4.6.A.1
6	D	6.F	5.1.A	5.1.A.3
7	B	5.F	6.4.A	6.4.A.1
8	D	5.F	7.7.A	7.7.A.1
9	B	5.F	7.9.A	7.9.A.1
10	C	5.F	7.12.A	7.12.A.1
11	C	5.F	8.1.A	8.1.A.4
12	B	5.F	9.9.A	9.9.A.2
13	A	5.E	4.2.A	4.2.A.3
14	C	2.C	3.4.A	3.4.A.1
15	B	2.B	5.2.A	5.2.A.1

Free-Response Question	Question Type	Skill	Learning Objective
1	Short	3.A, 3.B, 6.D	1.6.A
2	Long	1.A, 3.B, 4.C, 5.C, 5.E, 5.F, 6.D	2.5.A, 2.7.A, 4.2.A, 4.5.A, 4.7.A, 6.9.B, 9.1.A, 9.3.A

The scoring information for the questions within this course and exam description, along with further exam resources, can be found on the **AP Chemistry Exam Page** on AP Central.

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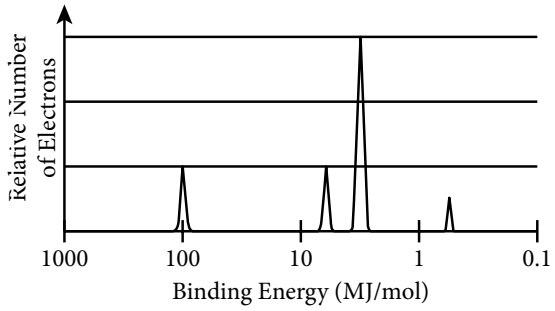
Scoring Guidelines

Question 1: Short-Answer



1. The complete photoelectron spectrum of an unknown element is given above.
   (a) Draw an X above the peak that corresponds to the orbital with electrons that are, on average, closest to the nucleus. Justify your answer in terms of Coulomb's law.
   (b) Based on the spectrum, write the complete electron configuration of the element.
   (c) On the graph, draw the peak(s) corresponding to the valence electrons of the element that has one more proton in its nucleus than the unknown element has.

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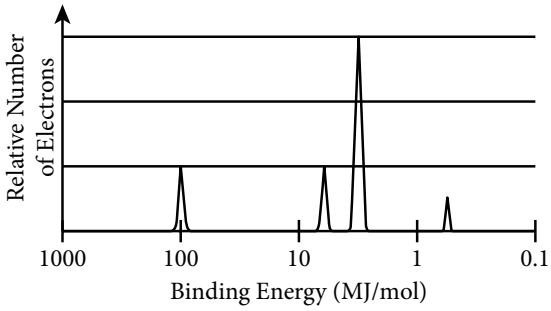
Scoring Guidelines for Question 1: Short-Answer — 4 points

Learning Objectives: 1.6.A

(a) Draw an X above the peak that corresponds to the orbital with electrons that are, on average, closest to the nucleus. Justify your answer in terms of Coulomb's law. — **1 point** 6.D 1.6.A



See the student's drawing.

One point for the X located above the leftmost peak (at 100 MJ/mol) in the spectrum with the justification that the electrons closest to the nucleus have the greatest binding energy because the strength of attraction between the charges (electron and nucleus) is greatest when the distance between them (r) is the least.

(b) Based on the spectrum, write the complete electron configuration of the element. — **1 point** 3.B 1.6.A

One point for the correct configuration.

$1s^2\ 2s^2\ 2p^6\ 3s^1$

(c) On the graph, draw the peak(s) corresponding to the valence electrons of the element that has one more proton in its nucleus than the unknown element has. — **1 point** 3.A 1.6.A

See the student's drawing.

One point if the peak is located just to the left of the rightmost peak in the spectrum.

One point if the height of the peak is twice the height of the rightmost peak in the spectrum. — **1 point** 3.A 1.6.A

Total for part (c) — **2 points**

Total for question 1 — **4 points**

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Question 2: Long-Answer

2. The following questions relate to sulfur and some of its compounds.

(a) Write the balanced equation for the combustion of $S_8(s)$ to form $SO_2(g)$.

(b) Calculate the volume of $O_2(g)$, measured at 1.00 atm and 298 K, that is required to completely combust a 500.0 g sample of pure $S_8(s)$.

(c) A student claims that the combustion of S_8 is an oxidation-reduction reaction. Justify the claim by identifying the oxidation numbers of sulfur and oxygen both before and after the reaction.

(d) In the box below, draw a Lewis electron-dot diagram for one valid resonance structure of SO_2.

(e) Based on the diagram you drew in part (d), what is the approximate oxygen-sulfur-oxygen bond angle in SO_2?

SO_2 can be oxidized to form SO_3 according to the following equation.

$$2\ SO_2(g) + O_2(g) \rightarrow 2\ SO_3(g) \quad \Delta H^\circ = -198\ \text{kJ/mol}_{rxn}$$

(f) Is the value of ΔS° for the reaction represented above positive or negative? Justify your answer.

$$2\ SO_2(g) + O_2(g) \rightarrow 2\ SO_3(l)$$

(g) Is the magnitude of ΔH° for the reaction to form $SO_3(l)$, represented above, greater than, less than, or equal to the magnitude of ΔH° for the reaction to form $SO_3(g)$? Justify your answer.

(h) Based on the information above, how does the thermodynamic favorability of the reaction change as the temperature of the reaction system is decreased? Justify your answer.

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Scoring Guidelines for Question 2: Long-Answer — 10 points

Learning Objectives: 2.5.A 2.7.A 4.2.A 4.5.A 4.7.A 6.9.B 9.1.A 9.3.A

(a) Write the balanced equation for the combustion of $S_8(s)$ to form $SO_2(g)$. — **1 point** 5.E 4.2.A

One point for the balanced equation.

$S_8(s) + 8\ O_2(g) \rightarrow 8\ SO_2(g)$

(b) Calculate the volume of $O_2(g)$, measured at 1.00 atm and 298 K, that is required to completely combust a 500.0 g sample of pure $S_8(s)$. — **1 point** 5.F 4.5.A

One point for calculating the moles of O_2.

$$500.0\ g\ S_8 \times \frac{1\ \text{mol}\ S_8}{256.5\ g\ S_8} \times \frac{8\ \text{mol}\ O_2}{1\ \text{mol}\ S_8} = 15.60\ \text{mol}\ O_2$$

One point for calculating the volume of O_2. — **1 point** 5.F 4.5.A

$$V = \frac{nRT}{P} = \frac{(15.60\ \text{mol})(0.08206\ \text{L}\cdot\text{atm}\cdot\text{mol}^{-1}\cdot\text{K}^{-1})(298\ \text{K})}{1.00\ \text{atm}} = 381\ \text{L}$$

Total for part (b) — **2 points**

(c) A student claims that the combustion of $S_8(s)$ is an oxidation-reduction reaction. Justify the claim by identifying the oxidation numbers of sulfur and oxygen both before and after the reaction. — **1 point** 1.A 4.7.A

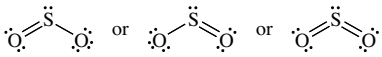
One point for all four correct oxidation numbers.

Oxidation numbers before the reaction: S = 0, O = 0

Oxidation numbers after the reaction: S = +4, O = –2

(d) In the box below, draw a complete Lewis electron-dot diagram for one valid resonance structure of $SO_2(g)$. — **1 point** 3.B 2.5.A

Any one of the three following diagrams is acceptable.



One point for the correct number of electrons.

One point for a valid Lewis diagram. — **1 point** 3.B 2.5.A

Total for part (d) — **2 points**

(e) Based on the diagram you drew in part (d), what is the approximate oxygen-sulfur-oxygen bond angle in SO_2? — **1 point** 1.A 2.7.A

One point for an angle that is consistent with the student's Lewis structure: 120°

(f) SO_2 can be oxidized to form SO_3 according to the following equation. — **1 point** 4.C 9.1.A

$2\ SO_2(g) + O_2(g) \rightarrow 2\ SO_3(g) \quad \Delta H° = -198\ \text{kJ/mol}_{rxn}$

Is the value of $\Delta S°$ for the reaction represented above positive or negative? Justify your answer.

One point for indicating negative with a valid explanation that mentions the smaller number of moles of gas in the products.

- *Negative, because the reactants are three moles of gas but the products are only two moles of gas.*

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(g) $2\ SO_2(g) + O_2(g) \rightarrow 2\ SO_3(l)$

1 point

6.D

6.9.B

Is the magnitude of $\Delta H°$ for the reaction to form $SO_3(l)$, represented above, greater than, less than, or equal to the magnitude of $\Delta H°$ for the reaction to form $SO_3(g)$? Justify your answer.

One point for indicating greater magnitude with a valid justification:.

- *Greater, because the enthalpy of $SO_3(l)$ is lower than the enthalpy of $SO_3(g)$ (by an amount equal to the enthalpy of vaporization of $SO_3(l)$), which makes the difference between the enthalpy of the reactants and the enthalpy of the products a larger amount.*

(h) Based on the information above, how does the thermodynamic favorability of the reaction change as the temperature of the reaction system is decreased? Justify your answer.

1 point

5.C

9.3.A

One point for indicating an increased thermodynamic favorability along with a valid justification.

- *$\Delta G_{rxn} = \Delta H_{rxn} - T\Delta S_{rxn}$. Assuming that both ΔH_{rxn} and ΔS_{rxn} are constant, as the value of T is decreased, the smaller in value the term ($T\Delta S_{rxn}$) becomes, making the term ($\Delta H_{rxn} - T\Delta S_{rxn}$) more negative. Thus ΔG_{rxn} becomes more negative, increasing the thermodynamic favorability of the reaction.*

Total for question 2 **10 points**

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